Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D

Experiment No. 04

Aim: To understand Continuous integration, Jenkins installation**.**

LO mapped: LO1, LO3

Theory: Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

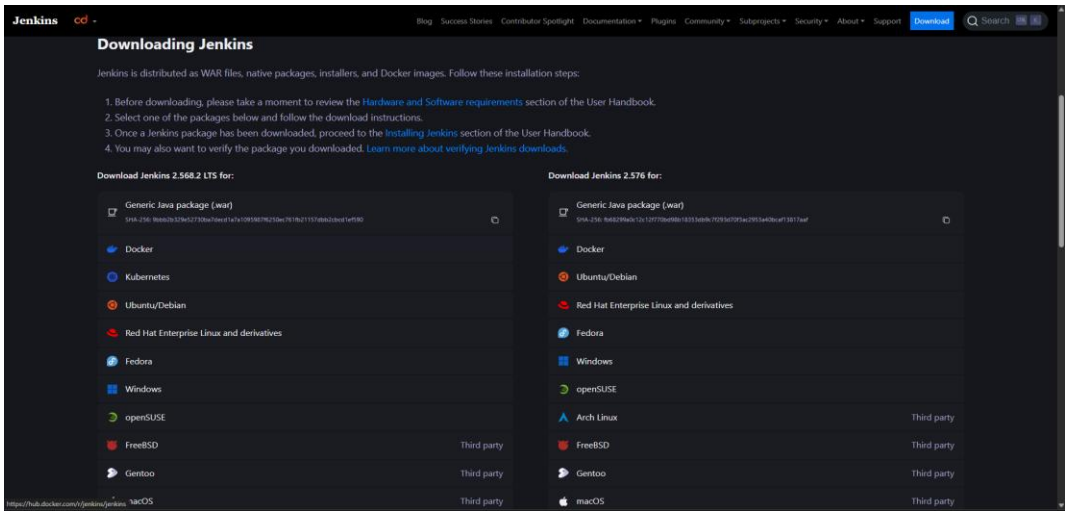
Jenkins is an open-source automation server that facilitates continuous integration and continuous delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supportsa wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

Steps to be followed while installing Jenkins:

1. **Java Installation:** Ensure Java Development Kit (JDK) is installed on your system**.**

```
C:\Users\203>java --version
java 25 2025-09-16 LTS
Java(TM) SE Runtime Environment (build 25+37-LTS-3491)
Java HotSpot(TM) 64-Bit Server VM (build 25+37-LTS-3491, mixed mode, sharing)
```

2. **Download Jenkins:** Get the latest Jenkins WAR file from the official website or package manager.



3. **Launch Terminal**: Open a terminal/command prompt on your system.

Name: Gunjan Rupani
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```
C:\Users\Lab203>java -jar "C:\Users\Lab203\Downloads\jenkins (1).war"
Running from: C:\Users\Lab203\Downloads\jenkins (1).war
webroot: C:\Users\Lab203\.jenkins\war
2026-08-06 02:25:06.558+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-06 02:25:07.418+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-06 02:25:07.457+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 25.0.2+10-LTS-6
2026-08-06 02:25:07.909+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-06 02:25:07.941+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Users/Lab203/.jenkins/war/WEB-INF/lib/xstream-1.4.21.j
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-06 02:25:08.273+0000 [id=3]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\Lab203\.jenkins found at: $user.home/.jenkins
2026-08-06 02:25:08.361+0000 [id=3]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@2a9bc08f{Jenkins v2.568.2,/,b=file:///C:/Users/Lab203/.jenkins/wa
AILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@50f40653{STARTED}}
2026-08-06 02:25:08.384+0000 [id=3]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@14379273{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-06 02:25:08.385+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@7918c7f8{STARTING}[12.1.8,sto=0] @930ms
2026-08-06 02:25:08.385+0000 [id=67]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-06 02:25:08.449+0000 [id=60]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-06 02:25:08.508+0000 [id=73]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgrading Jenkins. The last running version was 2.568.1. This Jenkins is version 2.568.2.
2026-08-06 02:25:08.508+0000 [id=74]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-06 02:25:08.547+0000 [id=73]    INFO    hudson.PluginManager#loadDetachedPlugins: Upgraded Jenkins from version 2.568.1 to version 2.568.2. Loaded detached plugins (and dependencies): []
2026-08-06 02:25:08.698+0000 [id=108]   INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-06 02:25:10.719+0000 [id=96]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-06 02:25:10.728+0000 [id=73]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-06 02:25:10.729+0000 [id=105]   INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-06 02:25:10.835+0000 [id=111]   INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-06 02:25:11.253+0000 [id=105]   INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-06 02:25:11.253+0000 [id=75]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-06 02:25:11.273+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-06 02:25:11.279+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-06 02:25:11.289+0000 [id=127]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-06 02:25:11.291+0000 [id=84]    INFO    j.install.InstallState$Upgrade#applyForcedChanges: New API token system configured with insecure options to keep legacy behavior
2026-08-06 02:25:11.305+0000 [id=85]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-06 02:25:11.327+0000 [id=60]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
WARNING: A restricted method in java.lang.System has been called
```

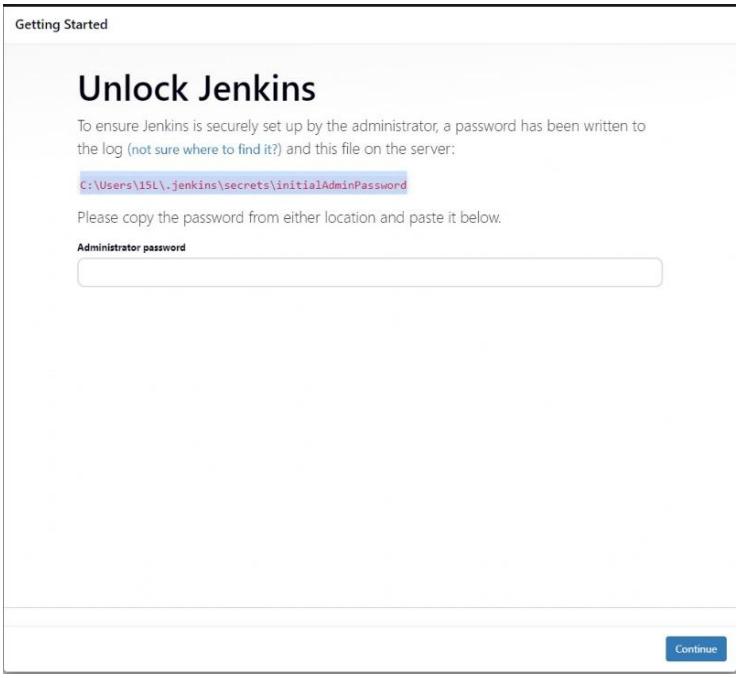
4. **Navigate to Directory**: Use the cd command to navigate to the directory containing the Jenkins WAR file.

```
2026-08-06 02:43:43.100+0000 [id=76]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-06 02:43:43.109+0000 [id=97]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-06 02:43:43.617+0000 [id=104]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-06 02:43:43.621+0000 [id=107]   INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-06 02:43:43.622+0000 [id=107]   INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-06 02:43:43.712+0000 [id=90]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-06 02:43:43.713+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-06 02:43:43.713+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-06 02:43:43.714+0000 [id=95]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-06 02:43:43.726+0000 [id=127]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-06 02:43:43.892+0000 [id=76]    INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 3abf6ea7d2c54abba3a9c764f1119839
[CRLF]>
[CRLF]> This may also be found at: C:\Users\Lab203\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************

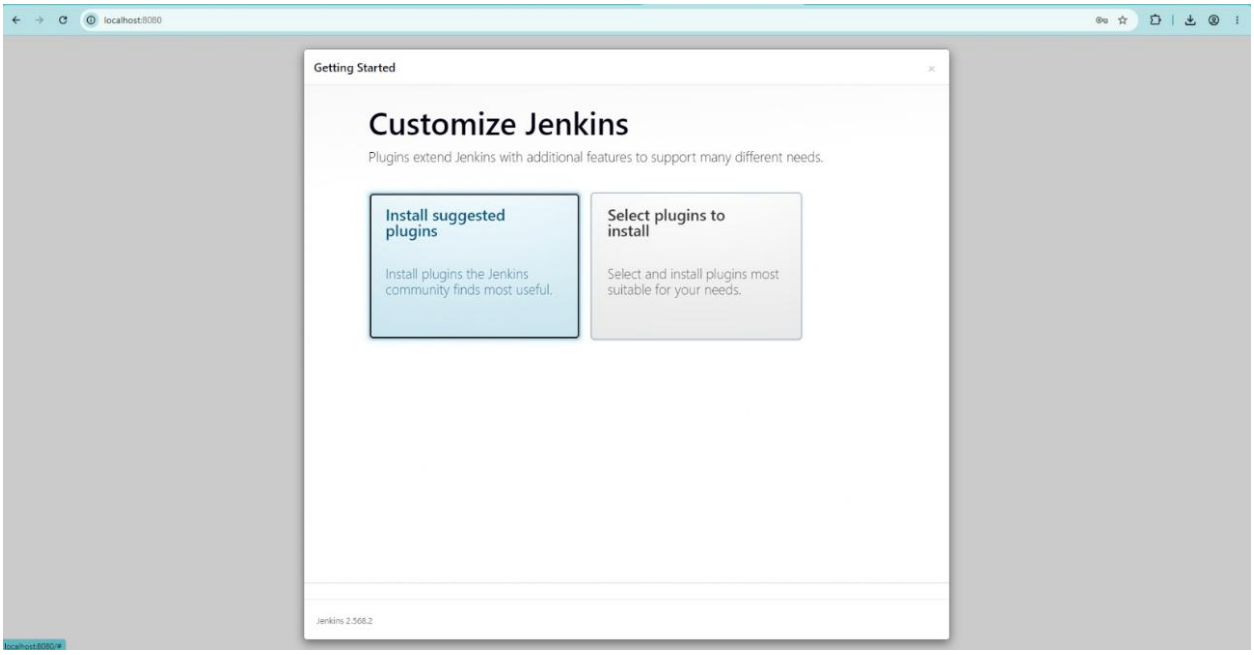
2026-08-06 02:43:47.289+0000 [id=76]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-06 02:43:47.308+0000 [id=60]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-06 02:43:49.134+0000 [id=127]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-08-06 02:43:49.134+0000 [id=127]   INFO    hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.jna.Native in an unnamed module (file:/C:/Users/Lab203/.jenkins/war/WEB-INF/lib/jna-5.18.1.jar)
```

5. **Start Jenkins**: Run the command java -jar jenkins.war to start the Jenkins server.
6. **Access Web UI**: Open a web browser and go to http://localhost:8080 to access the Jenkins setup wizard.

Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D

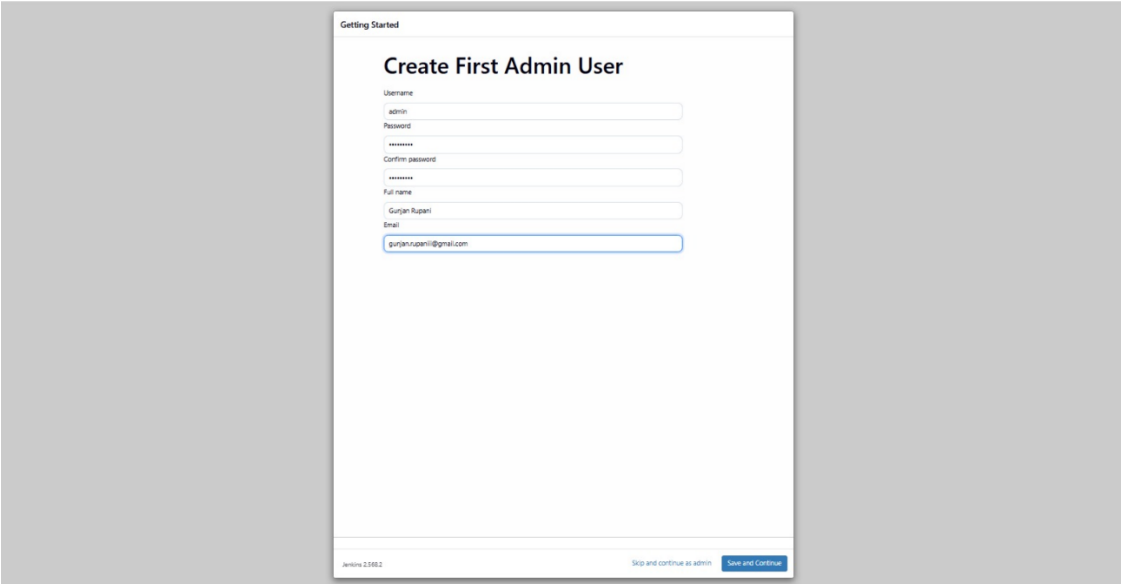


7. **Unlock Jenkins**: Retrieve the initial admin password from the terminal and enter it in the setup wizard.
8. **Customize Plugins**: Choose recommended or custom plugins to install during setup.



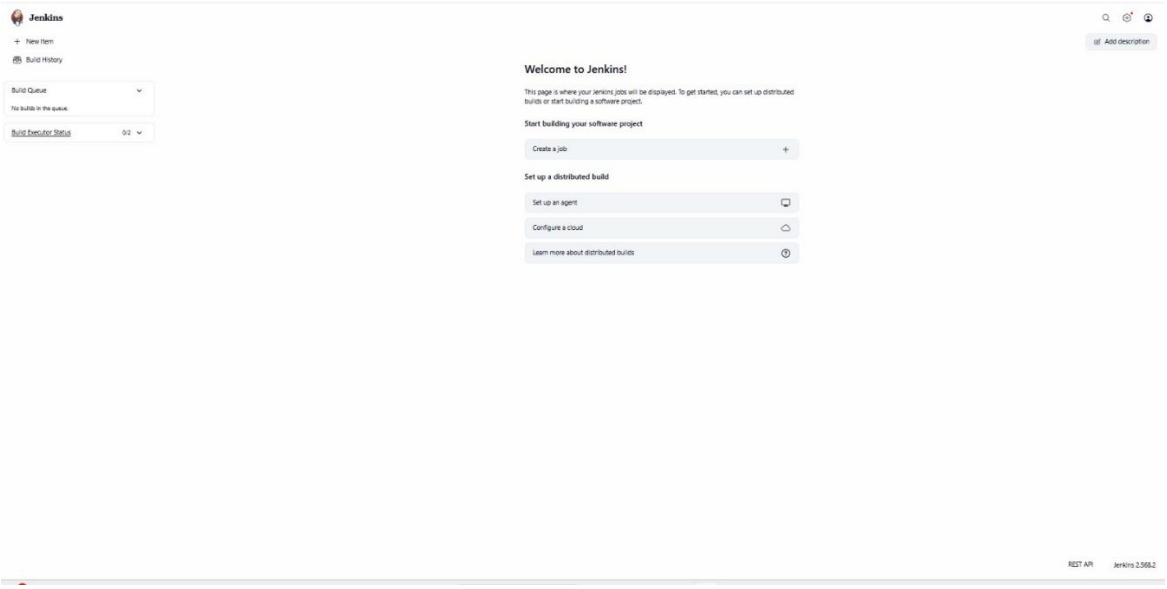
9. **Create Admin User**: Set up an admin user account with credentials.

Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D



10. **Jenkins URL**: Confirm the Jenkins URL and complete the setup process.

11. **Dashboard**: Log in with the admin credentials to access the Jenkins dashboard.



Conclusion: By this assignment we learned about Continuous integration and way to installJenkins.

Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D

AIM: To build pipeline using Jenkins.

THEORY:

What is a Pipeline in Jenkins?
A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a continuous integration/continuous delivery (CI/CD) environment.
In simple terms, it's a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline:

• Automates CI/CD processes
Provides better visibility into the build process
• Offers version control of build logic (Pipeline as Code)
• Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines:

There are two main types of syntax used to define Jenkins pipelines:

1. Declarative Pipeline

- Simpler and more structured syntax
- Designed for users who want an easy-to-read format
- Introduced later to make pipelines more readable and maintainable

Key Features:

Uses pipeline {} block
Organized into stages and steps
Easier error handling
Better validation in Jenkins UI

2. Scripted Pipeline

- More flexible and powerful

Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D

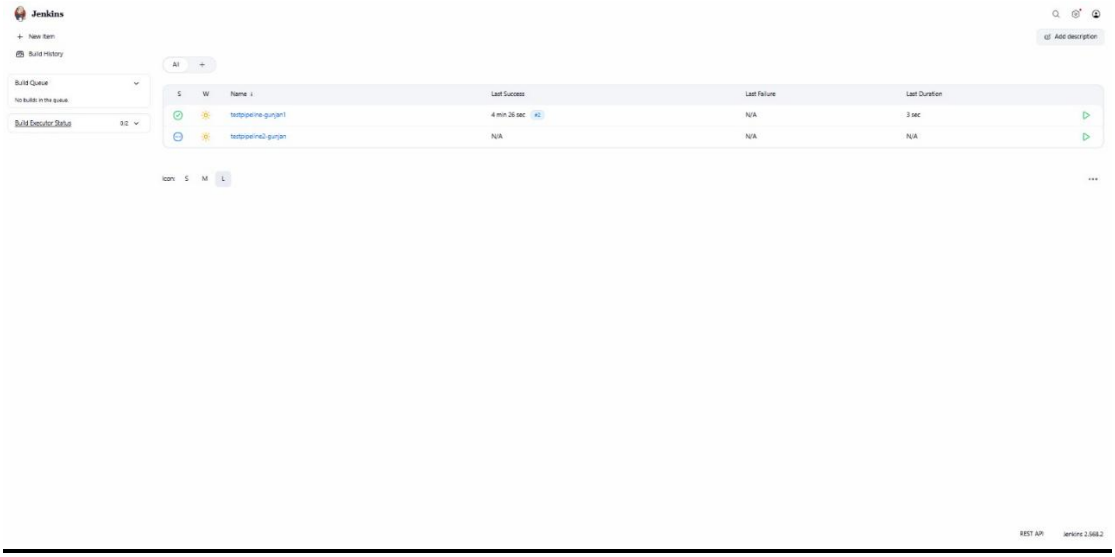
- Written in Groovy-based syntax
- Gives full control over the build process
- Suitable for advanced users

Key Features:

- Uses node {} block
- Everything is written as code
- Allows complex scripting and logic
- Less user-friendly for beginners

1.Declarative Pipeline

CREATION:

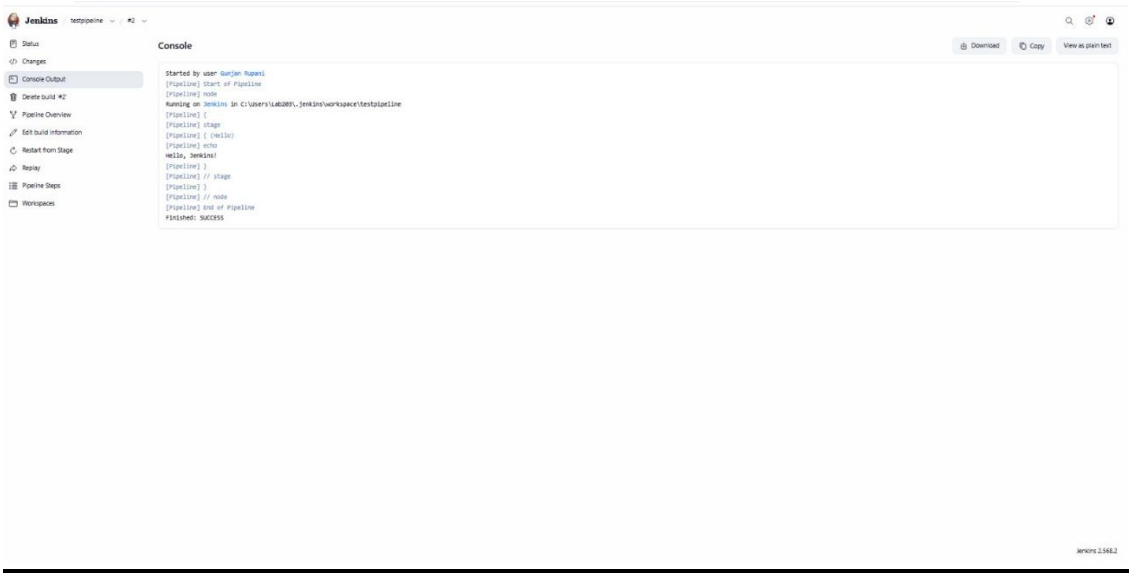


CODE:

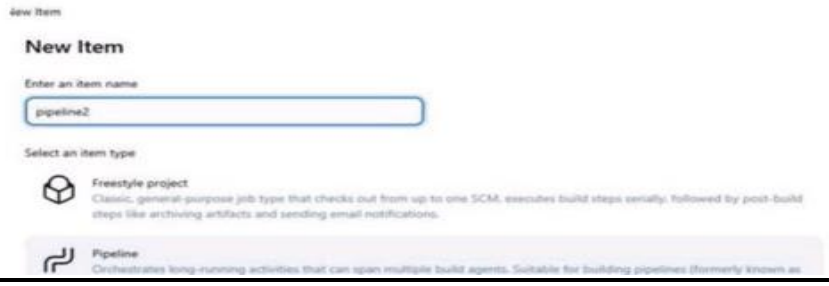
```
pipeline {
    agent any
    stages {
        stage('Hello') {
            steps {
                echo 'Hello, Jenkins!'
            }
        }
    }
}
```

OUTPUT:

Name: Gunjan Rupani
Batch: T22
Roll No.: -2401096
Subject:ASD&D



2.SCRIPTED CREATION:



CONCLUSION: LO1, LO3 Inthis assignment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.